



STUDENT SPRING CONFERENCE

INTEGRATING CTGAN AND KAN MODELS IN CREDIT SCORING

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OUTLINE



**ANALYSIS OF
PROBLEM**



**CURRENT STATE OF
THE ART MODEL**



OUR PROPOSAL



RESULTS



IMPACT



ANALYSIS OF PROBLEM





ANALYSIS OF PROBLEM

REGULATORY RESTRICTIONS 



ANALYSIS OF PROBLEM

REGULATORY RESTRICTIONS

SEVERE CLASS IMBALANCE

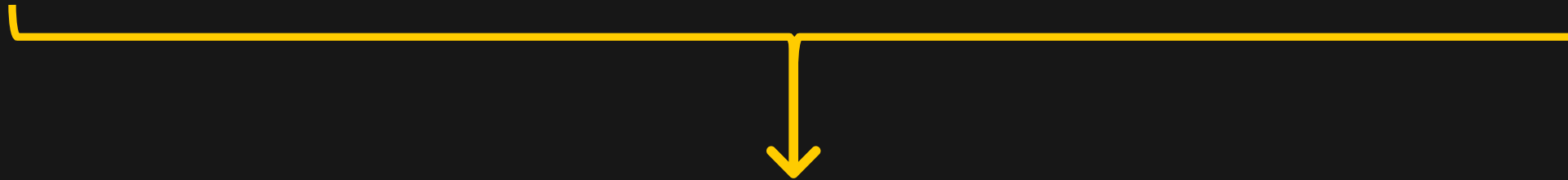




ANALYSIS OF PROBLEM

REGULATORY RESTRICTIONS

SEVERE CLASS IMBALANCE



PERSISTENT BOTTLENECK

(**not** straightforward obtaining large data volumes)



CURRENT STATE OF THE ART MODEL



CTGAN

Old solution (2019)



CURRENT STATE OF THE ART MODEL



CTGAN

Old solution (2019)

GENERATOR

**GENERATING
ARTIFICIAL DATASET**

DISCRIMINATOR

**DETECTING IMPERFECTIONS IN
GENERATED DATA**



CURRENT STATE OF THE ART MODEL



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Old solution (2019)

GENERATOR

DISCRIMINATOR

 GENERATING...



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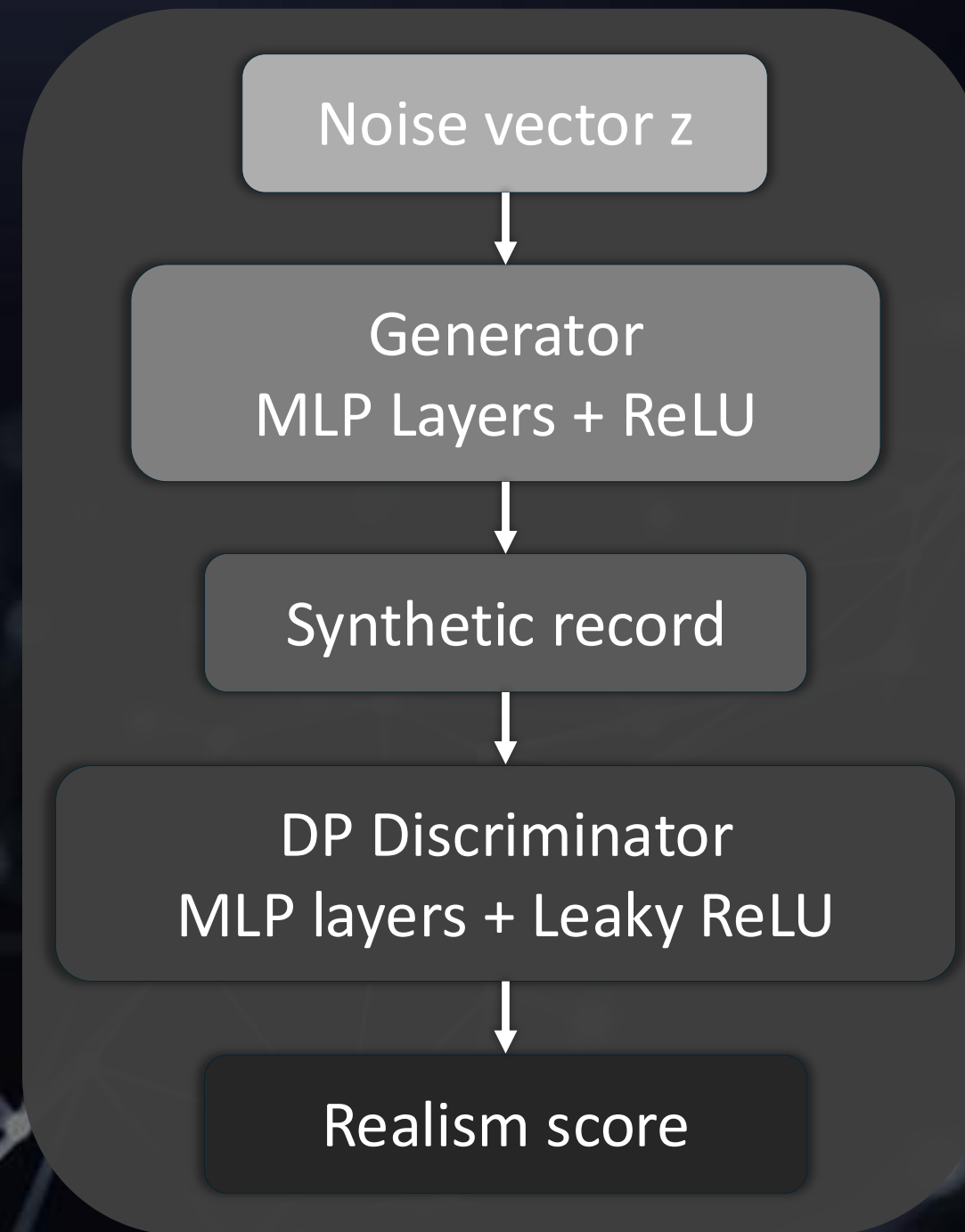




CURRENT STATE OF THE ART MODEL

ARCHITECTURE CTGAN

Old solution (2019)





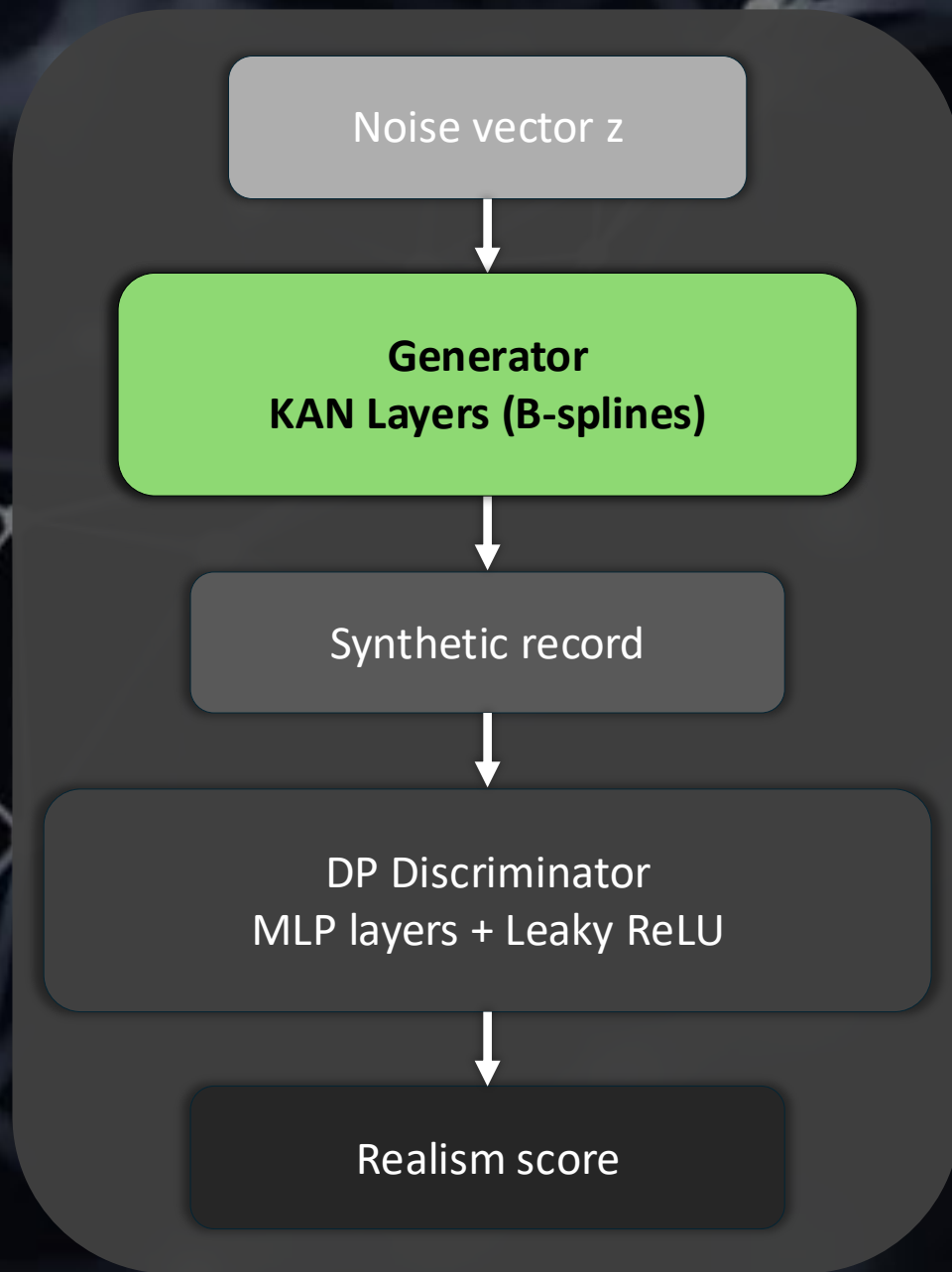
OUR PROPOSAL

KAN-CTGAN-DP



OUR PROPOSAL

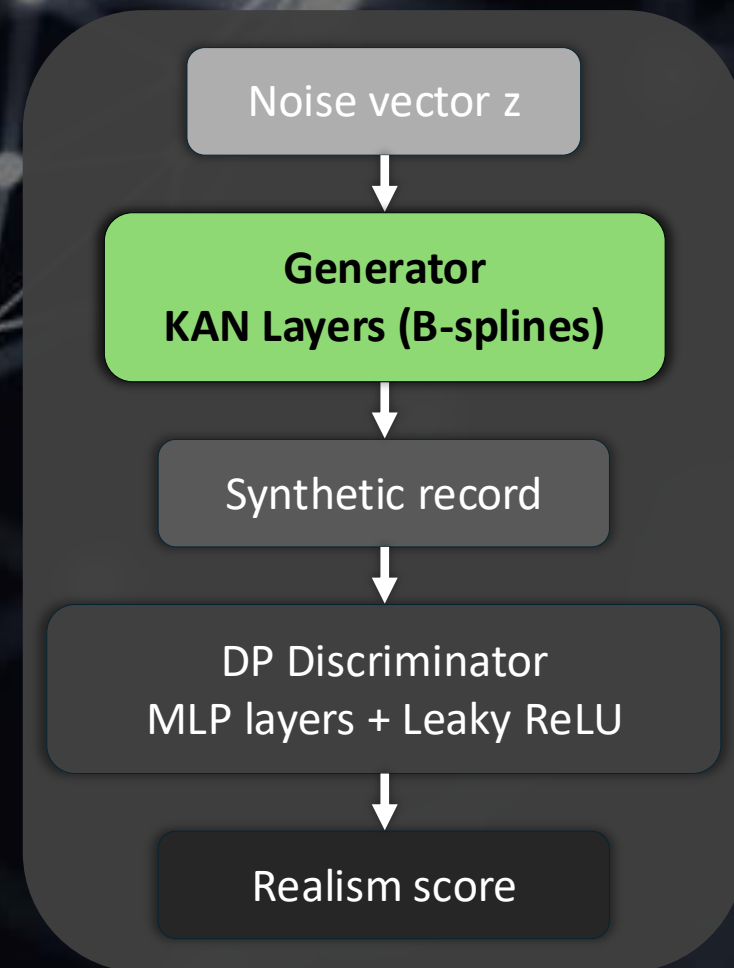
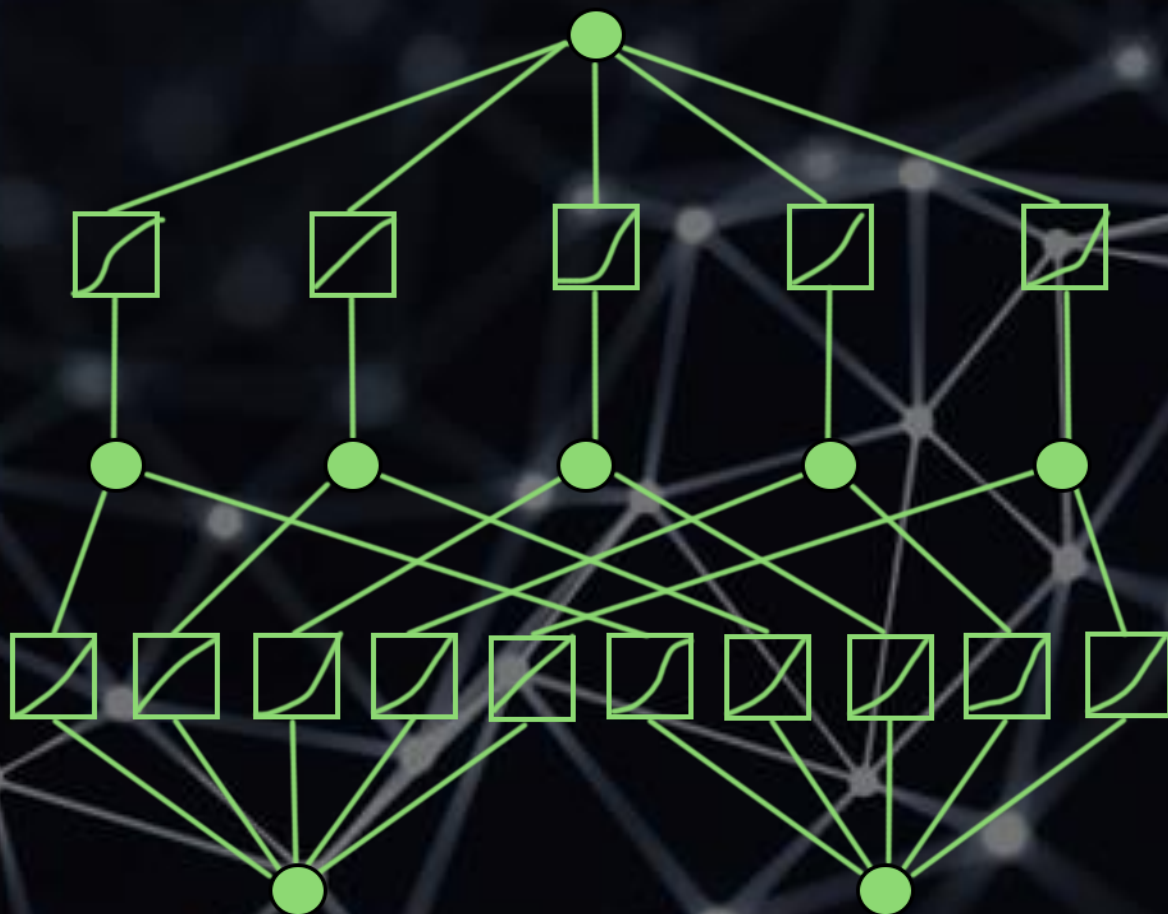
KAN-CTGAN-DP





OUR PROPOSAL

KAN-CTGAN-DP



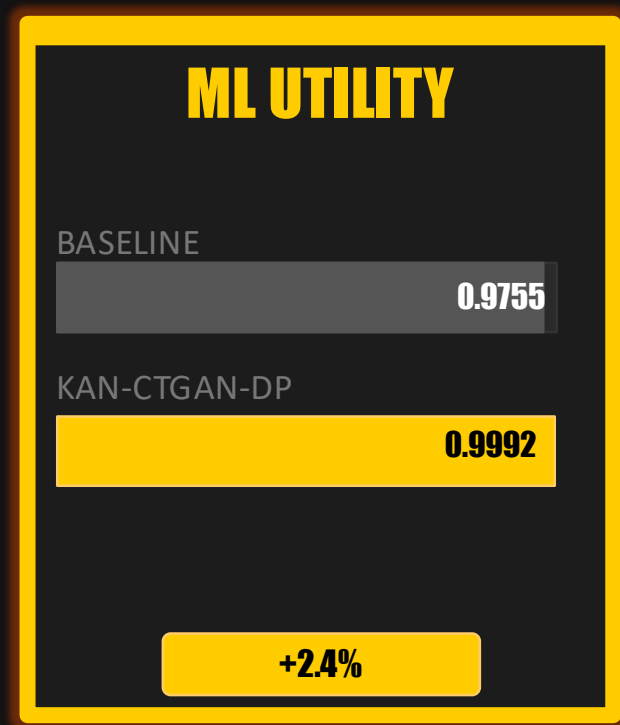


RESULTS





RESULTS





RESULTS

ML UTILITY

BASELINE

0.9755

KAN-CTGAN-DP

0.9992

+2.4%

PRIVACY

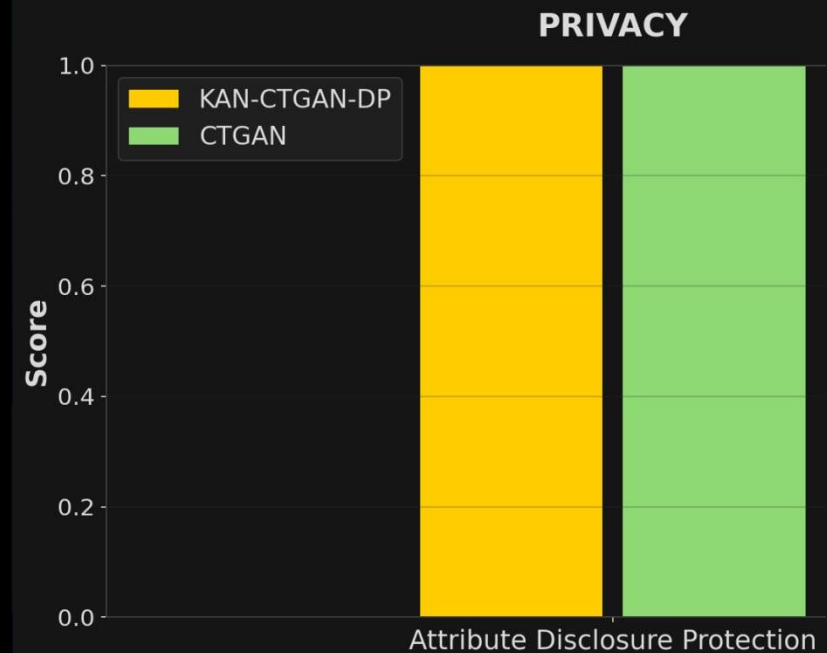
BASELINE

0.9980

KAN-CTGAN-DP

0.9975

≈ 0%



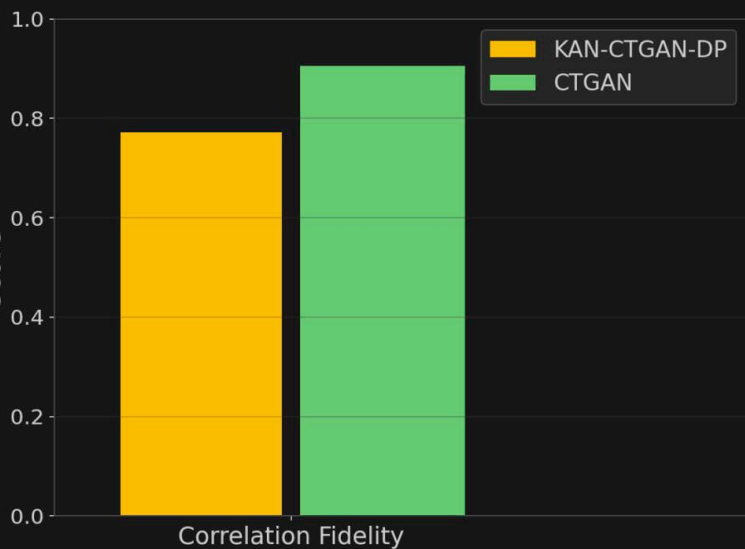


FIDELITY





FIDELITY



FIDELITY

CORRELATION FIDELITY

BASELINE

0.8306

KAN-CTGAN-DP

0.7782

-6.3%



IMPACT





POINT 1

**BETTER SYNTHETIC DATA QUALITY —
NO PRIVACY TRADE-OFF**





POINT 2

**MORE CREDIBLE DATA → MORE ACCURATE
MODELS → MINIMIZES BANK EXPENSES**



POINT 3

OPEN-SOURCE AND EXTENSIBLE TO OTHER HIGH-STAKES DOMAINS

GitHub:



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**YOUR ATTENTION IS
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